UNIVERSITY of DENVER

VLSI

October 9th, 2026

Due date: October 16th, 2026

Name:_____ DU ID:_____ Section:____

Lab 4 – Directions

Draft schematics, layouts, and symbols for two inverters having sizes of:

- 12u/6u (PMOS) and 6u/6u (NMOS)
- 48u/6u (PMOS) and 24u/6u (NMOS), where the devices use a multiplier, M = 4

Your lab report should document the following:

SCHEMATIC

- The schematics for the inverters and respective simulations to show the switching point.
- Apply a pulse waveform to both inverters each driving a 100 fF, 1pF, 10pF, and 100 pF capacitive load. Comment the results on the report.
- All necessary explanations to support all the decisions taken during the design of the schematics.

LAYOUT

- The layouts for the inverters
- The power should run on the top of the cell via metal1 and ground should run on the bottom of the cell also via metal1
- All necessary explanations to support all the decisions taken during the design of the layouts.

Good luck and have fun.